

APICULTURE

Honey bees are a group of polymorphic colonial insects. They produce nutritive honey and wax; both the products are of commercial value. From very early times man knew that bees produce delicious honey, that they sting, that they increase in number by swarming and that with smoke he can control honey-bees. It was during the early part of the 19th century many scientific studies were conducted which revealed the biology, behaviour, life cycle and development of honey bees.

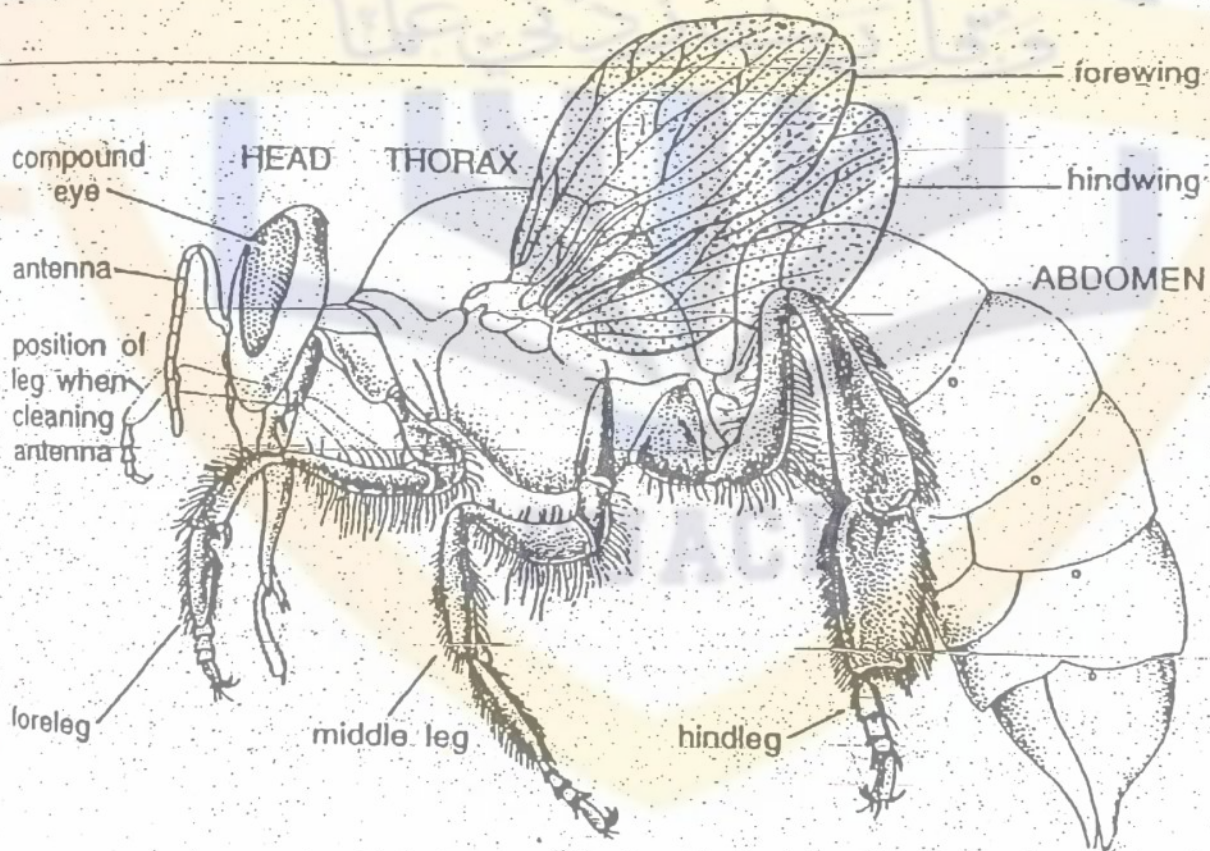


Fig. 3.1. Worker honey bee (lateral view).

Previously man extracted honey by destroying the whole bee colony and the honey extracted was also of low quality because of the crude and unscientific way of handling. Today with the treasure of knowledge he has acquired he is able to divide a colony to start and rear new colonies. Now sophisticated tools and new methods are available to extract honey from hives without much disturbing the hive colony and the wax combs can be reused. Pathology of honey bees are extensively studied and drugs are now available to control diseases in bees. Another leap in bee-culture technique is the artificial insemination of queens for increased honey production in colonies.

Honey bees are most familiar insects. They are best known for their labour, small saving, colonial life and division of labour. They are characterized by the presence of a highly specialized type of legs, mouth parts, membranous wings and communal behaviours. The "waggle dance" and "language of bee" is a remarkable feature. Preparation of chambered nest its hygiene and maintenance are points to be noted.

They belongs to genus *Apis* and a variety of species are found in it. Some of them are given below:

***Apis dorsata*-F. (Rock bee)**

This is the largest honey bee, about 2 cm. in length. These are the strongest and the most powerful amongst the bees. They make their combs in open areas like big branches of trees, rocks, old forts and other such places where man cannot reach easily. These are almost similar in size and not much of difference is noted amongst the queen and worker but for the colour of the abdomen, which is much darker in queen. The comb cells are almost of uniform size and a comb can store about 20-25 kg of honey. These bees are much irritable and ferocious so their domestication is not very easy.

***Apis Indica*-F. (Indian bee)**

These are smaller in size and brighter in colour than the *Apis dorsata*. These bees can be well marked by their nests as the nests are always in dark and protected places, like old forts, caves and crevices, hollow trunk of trees etc. Several combs are made parallel to each other but honey is found in only one or two combs only. Total production per comb per year is not more than 2-6 kg. under favourable conditions. These are found in hills and plains and can easily be domesticated because of their mild nature.

Apis Floria - F. (Little bee)

These are smallest and most gentle bees which make their nest in bushes, trees houses and other surrounding. They are docile and gentle in nature and can be domesticated easily but yield is much less and uneconomic. The comb can be removed easily and honey can be extracted without much efforts.

A. mellifera - F. (European bee)

A variety of European bees are found in general but the Italian species proves to be the best. The nature of these bees resembles with that of *A. indica*. These too make several combs parallel to each other and brood is in the central comb. The amount of honey is higher and the nature of bee is gentle. These can be domesticated and their variety can be easily improved by breeding. The queens more prolific and honey gathered varies from 5-10 kg. per comb per year. They are common in European countries.

Melipona and Trigona - Spp. (Dammer bee)

These are different from the above said honey bees. Much smaller in size and are unable to secrete wax for making the comb. They make the comb by using earthen mixture, sand and resinous substances from plants. The comb thus is a very tiny comb with 100-150 gm. of honey. Such a comb can easily be seen in the shrub and gardens close to agricultural fields. They are best for pollination of the flower.

COLONIAL ORGANISATION AND DIVISION OF LABOUR

The honey bee presents an example of social and organized living. The member are well defined into various castes with their special works. They are divided into (1) Queen, (2) Drone and (3) Worker. Numbers of insects in a colony varies according to species of the bee and ranges from 40-80,000 per colony in extreme cases where as in case of *Melipona* or other it may be only few thousands only.

Queen

Largest of all the bees with small legs and weak wings. Queen bee is the real ruler of the hive. She has an extended abdomen, full of eggs and the last segment is modified as an ovipositor. She has no work except eating and laying eggs. She lives for 3-3½ years and lays about 14-15,00,000. eggs during her life span. The rate of egg laying under favourable conditions may be 2,000 eggs per day. Generally only one queen is founding in a comb but in case when her egg laying capacity is reduced she can be changed and replaced by another queen. In

extreme cases such a old queen comes out of the hive and establishes another hive. In some of the cases a colony may have more than one queen but only one is allowed to live in the hive rest are pressed to leave the nest.

Queen is supposed to be the mother of all the drones and workers and queens (if they are) present in the comb. She lays both fertilized and unfertilized eggs which later on develop as male and workers. Some of the workers which are in real sense the females are fed with a special substance called "Royal Jelly" and the larvae fed on royal jelly develops as a queen rest as workers. By the time the egg laying capacity of the queen goes slow a new larvae is fed with royal Jelly and is developed as future queen.

A single fertilization is sufficient for the whole life. The queen moves from cell to cell of the comb and lays eggs. She has no special cell to live in and is always surrounded by the workers. She is generally found in the lower border of the hive. A mated queen collects the sperms in a receptacle and it is on her will to decide which batch of the eggs should be fertilized and which should not the males or drones are generally unfertilized whereas the queen and the workers develop from the fertilized eggs. There is no difference in the organs or morphology of the worker or queen but for the size of legs and wings. The queen bee is fed with a highly nitrogenous food mixed with pollen honey and a secretion from the lateral pharyngeal glands. All these are mixed and make as Royal Jelly. An egg develops as a larva in 3 days and grub into pupa in 5-7 days. The pupa emerges out after a week as an adult.

Workers

More than 90% of the total population present in the comb consists of workers. These workers are the atrophid female which are unable to

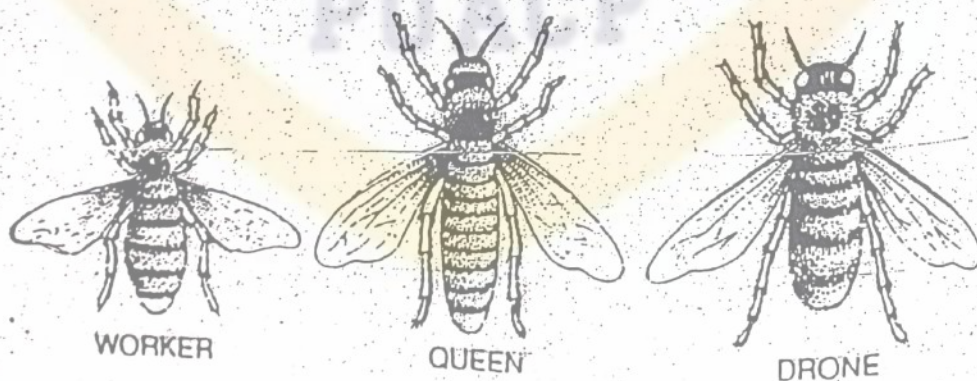


Fig. 3.2. Castes of hone bee.

lay eggs and they sacrifice their total life in the service of queen and the hive and its residents. The longevity of a worker depends upon the type of work and activity, generally they live for about three months in a off season but in busy seasons this is reduced to half.

The newly developed worker has weak wings, so she is employed to feed the grubs, ^{Caribal form} help the queen in laying eggs, cleaning the hive and they are named after their work as nurse-bee, cleaner-bee or builder-bee.

The builder bees are provided with a wax gland in their 4 to 7th abdominal segments on the ventral side and these are engaged in making the nest cells whereas some of the bees produce bee glue (propolis), which is used in repairing the cracks and crevices, there are known as repairers. The duties of a nurse bee is to produce Royal Jelly for the developing queen and "bee-bread" for rest of the grubs. She looks after the development of the grub and before they pupate she seals the door of the cell. Cleaning of cell, maintenance of hygienic conditions removal of debris and dead bodies are done by the cleaners.

The workers have other important job to maintain the temperature of the hive. At the time of production of honey a fairly good amount

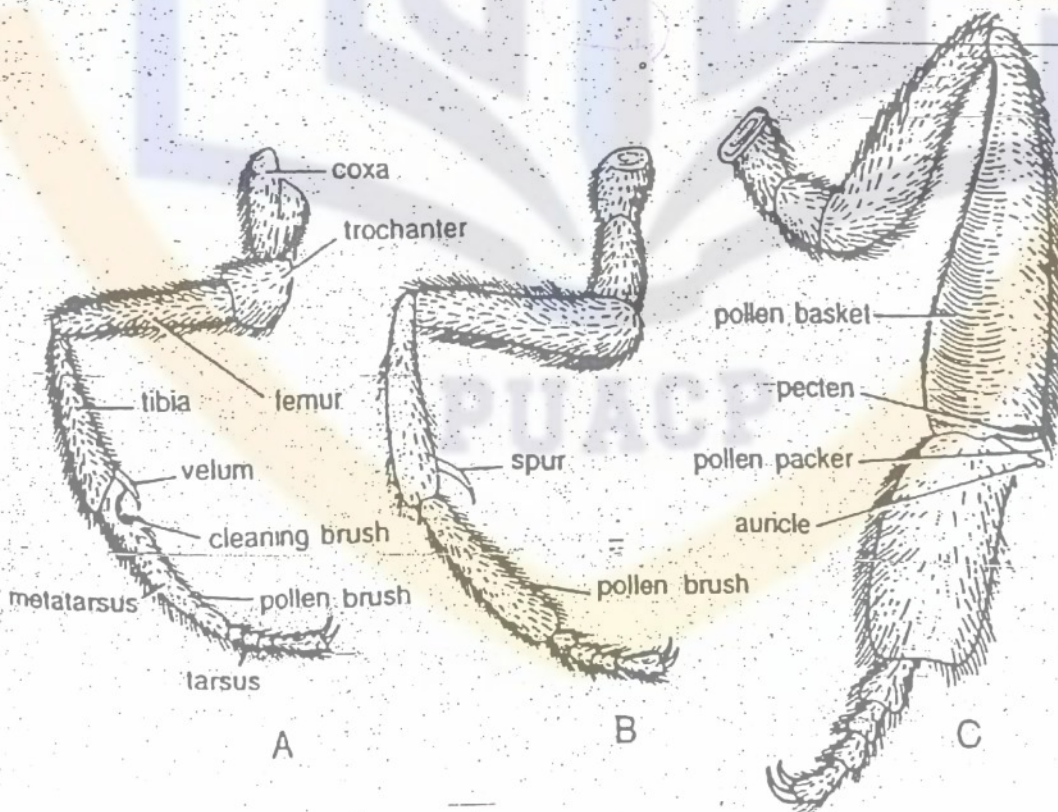


Fig. 3.3. Honey bee Legs. A-Prothoracic leg; B-Mesothoracic leg; C-Metathoracic leg.

of water is found which in later stage may spoil the honey. So to avoid the loss the worker bees assemble near the newly filled cell and deep that cell warm. The water vaporizes and level is lowered down. On reaching a certain point the bee seals the cell and protects the honey. On the other side the temperature may not damage the egg laying capacity of the female in hot summer she is always kept cool by the workers. There are certain bees with long wings they always vibrate their wings so the temperature is lowered down. In the month of June when the sun is too hot, some of the fanner bees go out and take a dip in water and again come near the queen. This gives an air conditioning to the queen.

In the same way they keep the queen warm during winter by assembling themselves near the queen.

Collection of food, pollen and honey is again an important work of the workers. These workers are good flier and can cover a long distance in a shorter duration. They have their own language and by dancing before other bee they tell the direction and distance of the food from the hive quantity of the food and its quality. Such bees are provided with long and powerful flight muscles and wings. Their legs are provided with pharyngeal legs. These legs are armed with big structure above the tibia called pollen basket, which helps in collecting the pollen grains.

Workers are the soldiers also they have a powerful sting in their last abdominal segment and protect the hive from outside attacks. They have a tendency to chase the enemies.

In terms of their body structure the workers are the most adapted to perform the duties in the colony. The adaptations are:

1. Well-developed proboscis for nectar collection.
2. Wax plates and wax glands are well formed for constructing and repairing the bee hive.
3. Very powerful sting to defend the colony.
4. Fully formed salivary and mandible glands honey stomach for jelly secretion.

Workers are the back bone of the colony. They carry out the routine chorus of the colony. The duties to be performed in the colony are divided among the worker bees based on their age. The major works they perform (in ascending order of age) are:

1. Cleaning the cells of the comb.
 2. Secretion of royal jelly and feeding the larvae and young ones.
- They are called 'nurse' bees.

3. Secretion of bees wax and construction and repair of honey combs.
4. Defend against intruders and protect the colony.
5. Collect pollen, nectar and water and keep the colony cool.
6. Act as scouts in locating new food sources and lead the members during swarming.

The life span of worker bee is 3 months to one year. During active seasons they have a short life span.

Drone

The males in the colony are called drone. They develop parthenogenetically and have no stings. They depend on workers for their own food and have the sole responsibility to fertilize the queen. They are of intermediate size and pupae period ranges from 10-13 days. Total adult life span ranges from 7-8 weeks. The reproductive organs are well developed and sometimes they die just after copulation. They are developed only at the time of breeding when a new queen is under process otherwise they are kicked out of the colony in day of scarcity. A hive generally keeps 200 drones at a time. Drone cells are little bigger than the workers cell and smaller than the queen cells.

Honey Comb

Various types of bees make various types of honey combs but their lay out pattern is almost same in all the cases. They have a common partition wall having serially arranged cells on either sides. These cells are hexagonal, their walled and fragile. The bee may have a separate comb as a brood chamber or may use the same hive as a brood chamber. In most of the cases the hive is divided into certain chambers the brood chamber having eggs larvae and pupae whereas storage is done in another chamber. The size of the cell for larvae and pupae of the brood would be male and workers differs in size. The cell of a drone is large as compared with that of the worker whereas that of queen is the largest. The storage cells are small and much more in number. These are built near the margin or on the top of the hive.

Life Cycle

The fertilized female lays eggs. Number of eggs laid daily vary according to the environmental condition but it may range from 1500-2000 eggs a day. The eggs are small, cylindrical, pale yellow or white. One egg per cell is laid by the female and eggs adhere with the cell wall. Generally three types of cells are found the Royal brood-cell, Drone brood cell and Ordinary brood cell which, give rise to queen, drone and the workers. The sex of the developing larva is decided by the quality and quantity of the food fed to the larvae. The eggs hatch

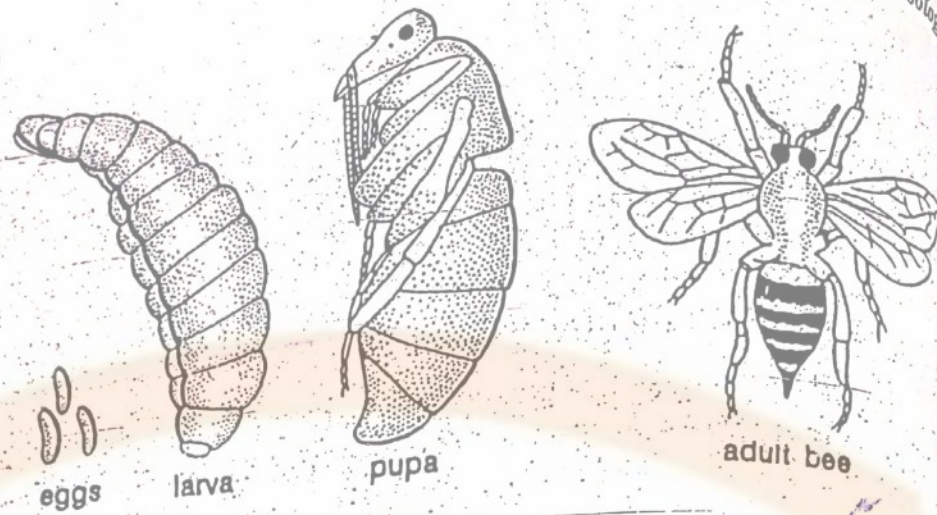


Fig. 3.4. Life history of *Apis indica*.

in three days giving out of the grubs. Grubs are yellowish or white and curved structure and develop in pupal form in 5-7 days their development depends on the food. Some of them are fed with Royal Jelly where others are fed with bee bread. Bee bread inhibits the growth of reproductive organs. So only workers come out of such larvae they pupate inside the cell and pupae come out as adult in 7-14 days. The queen, drone and workers have different life span.

	Egg	Larva	Pupa	Adult
Queen	3 days	5 days	7-8 days	15 years
Drone	3 days	7 days	14 days	56 days
Worker	3 days	4-5 days	11-12 days	56-70 days

Swarming

The process of leaving off the colony by the queen termed as swarming. It happens towards the end of spring or early summer but the real cause of swarming is still not well known. In summers when plenty of food is available and hive is overcrowded by the bees, the queen leaves the hive on a fine fore-noon with some old drones and workers and establishes a new colony at some other place. Now in the old hive a worker is given Royal Jelly and is converted into a new queen of the colony. This new empress of the colony never tolerates her successor, as a natural law in the hive. So she orders to kill the other sisters, if any, in the hive.

Supersedure

When the egg laying capacity of the old queen is lost or it suddenly dies, a new young and vigorous queen takes the position of the old queen and is called as supersedure.

Absconding

The migration of the complete colony from one place to another takes place due to some unfavourable conditions of life, such as destruction of the comb by termites or wax-moths and scarcity of nectar producing flowers around the hive. This phenomenon is quite different form that of swarming.

Nuptial or Marriage Fight

The first swarm is led by the old queen but the second swarm is led by the 7 days old virgin queen which is followed by the drones and is called marriage flight. One of the drones starts copulating with the queen in the sky and fertilizes the queen and dies during the course of copulation. The queen receives spermatophores and stores in the spermatheca. Along with the queen, died drone falls on the ground and the queen reaches the hive.

HIVE

The house of honey bees is termed as hive or comb. It consists of hexagonal cells made up of wax secreted by the worker's abdomen. These hives are hanging vertically from rock, building or branches of trees. Each hive has thousands of hexagonal thin walled fragile cells arranged in two opposite rows on a common base. The resins and gums secreted from the plants are used for the repairing of the hives. The young stages are generally occupying the lower and central cells

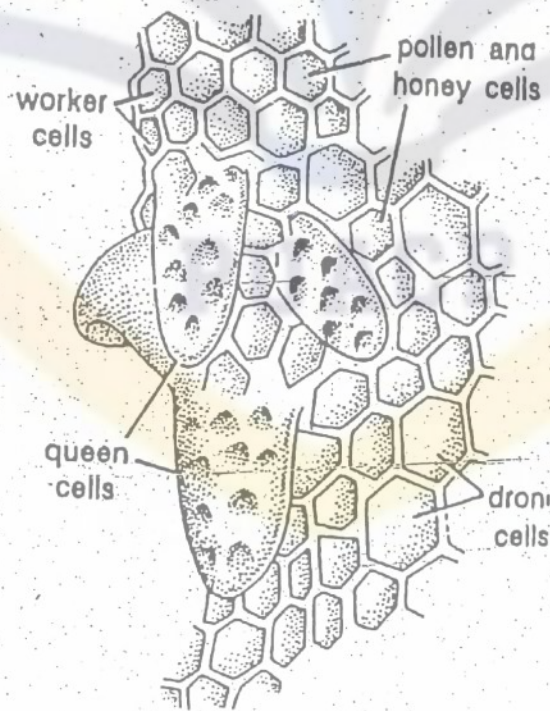


Fig 3.5. Hive of *Apis indica* showing various cells.

in the hive which are the 'Brood cells'. In *A. dorsata* brood cells, are similar in shape and size but in other species brood cells are of 3 types viz., *Worker cell* for workers, *Drone cell* for drones and *Queen cell* for the queen. The queen cell cannot be used again while the rest are used a number of times. There are no special cells for lodging the adults, which generally keep clustering or moving about on the surface of the comb. The cells are mainly intended for the storage of honey and pollen specially in the upper portion of the comb while those in lower part are for brood-rearing.

Honey

Honey is produced by the bee and is of an important medicinal value. It contains a variety of minerals e.g. Ca, Fe, Silica, Mg, Mn, K, Cl, S, Al, and phosphorus whereas amino acids, citric, formic, succinic, malic and acetic acids are also present in traces, pigments like Carotene, Xanthone and Anthocyanin and vitamin A, B complex and C are also reported in it. Besides all these enzymes like invertase, diastase are also found. The chemical composition of honey constitutes as follows:

Lavulose	38-41%
Dextrose	21-35%
Maltose and other sugars	8-10%
Enzymes and pigments	2-3%
Ash	1%
Water	17%

Pure honey gets granulated after long storage. The granulation becomes faster due to presents of air-bubble pollen, and other colloidal substance. It has density 1.2.

Economic Value

It is of great economic importance and is widely used in medicine and as antiseptic. It has its own food value and gives high calorific value and energy. It is used in candies, cakes, breads and cold drinks. Alcoholic drinks, livestock feeds and poison baits are also made from honey. It is the only substance which requires minimum energy for its digestion that is why it is given to new born infants.

Bees Wax

It is the most useful bi-products of the honey bee. It is secreted by the dermal glands of the abdomen from 4th to 7th segment and is completely soluble in ether, chloroform and other organic solvents.

Brown or yellowish brown in colour it does not dissolve in water. It melts easily.

It is used in manufacture of cosmetics creams, paints, ointments, plastic works, polish and a variety of lubricants.

Enemies of Bee

Honey being sweet, tasty it has a number of enemies also.

Wax Moth (*Galleria mellonella* Linn) lesser wax moth (*Achroia grisella*) are the worst enemies of honeybee. Their caterpillars creep inside the comb and feed upon the brood comb. They spoil the complete colony. Quite often the bees undergo absconding swarm and leave the hive.

Sphingid moth (*Achorontia styx*) is a strong and powerful moth, which enters the colony and feeds upon the honey, but the bees after sometime kill them.

Beside above mentioned insects black ants, termites, dragon flies are the powerful insects which destroy the bee hive.

Not only the invertebrates but some of the vertebrate pests like lizards, catoles, king crows, birds and bear cause serious damage to the bee hive and there is no escape for the bee.

The bees are subjected to bacterial diseases also like dysentery, fowl brood disease, caused by *Bacillus plutan* and *B. alvei*, Nosema and Bee paralysis. Certain mites like *Acarapis woodi* kill the bees and damage the hive.

SCIENCE OF BEE-KEEPING

Bee keeping is the art of rearing honey bees, to obtain honey for human use. The honey-man takes is actually excess left after the use by the bees. Other than a source of honey and bees wax, honey bees are good pollinators. (A thorough knowledge about the biology, behaviour, life history and development of honey bees must be obtained before venturing to 'Apiculture')

Wooden hives are made for keeping bees. Special tools are available these days like honey extruder, bee gloves smoker, and bee veil to protect the keeper. Artificial feeding method is also resorted to at time of food shortage.

Bee Hives

Man provides shelter space for the bees to set up colonies. This is called bee-hive. The hive space must be large enough for passage of bees and smaller enough to discourage comb building. In earlier days

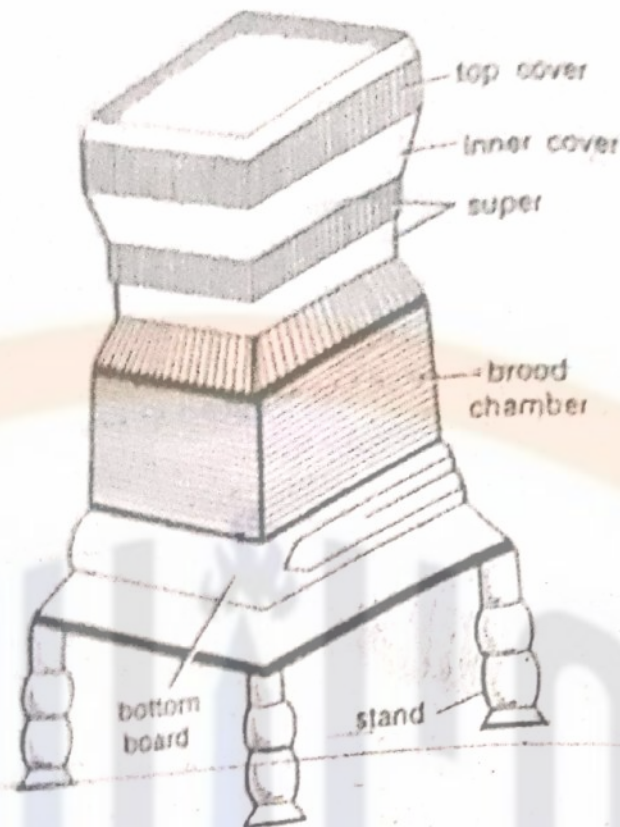


Fig. 3.6. Typical movable hive.

straw skep was used for bee keeping. Different models of hives were designed during the past years: the most rewarding was that of Langsworth. He designed a hive that can be opened at the top. The modern bee hive is the *movable-frame* hive. It is a wooden frame box made of single or double walls the former for warm place and the latter for cool climates. The hive consists of a *bottom board*, *brood chamber*, *super chamber* *inner cover* and *top cover*. They are placed one above the other and fixed on a stand. All these parts can be separated. The *bottom board* acts as the entrance for the bees. The brood chamber is a wooden box inside which numerous frames called "comb foundations" are fixed. Comb foundation consists of sheets of pure bee wax on which are embossed the base of cells of the honey comb. The worker bees secrete wax to extend the walls of these cells. The comb foundation helps in controlling the rise of the cells and reduce the number of drone cells. These frame of comb are movable and can be lifted, hence the name for the hive "*movable frame* hive."

The chamber above the brood chamber is the *super chamber* inside which honey is secreted and stored. Queen is prevented from entering this chamber by using *queen excluder*. This equipment has narrow spaces of 4 mm which allows only the workers to enter the super chamber.

The top cover can be lifted to inspect the state of the colony or honey formation. The wooden hive is coloured yellow or white on the outside for keeping the chambers cool and to aid easy recognition by the bees.

Other Appliances

Many tools are devised for proper care and maintenance of beehives. The bee veil is a device to protect the beekeeper from bee sting. The veil is made of fine nettings, with a string attached for tying. Gloves are sometimes used by keepers for protecting the hands. While inspecting the hives, smoker is an equipment to scare the bees. During hive maintenance and honey collection the worker bee may attack the keeper in groups. The smoker released will irritate the bees and keep them away.

Honey extractor is a metal drum inside which a rotor rotates in its axis. The foundation frame on which the bees secrete their honey, is fixed on to the rotor and rotated. This creates a centrifugal force, which separates the honey without disturbing the comb frame.

Other minor tools are bee brush for cleaning the combs and strapper for collecting excess honey.

Hive Maintenance and Management

To obtain good yield of honey the keeper must provide ideal conditions to the bees. Maintenance and management of bees there fore is very significant for the perpetuation of the colony. Following are the few important points of hive management.

1. The location of the hive should be such that the flora in the surrounding area and their flowering period should suit with the active period of the colony. The area must have good humidity, preferably protected from direct sunlight.
2. The hive may be single walled or double walled in structure based on the climatic features of the locality.
3. The keeper must be familiar with biology and behaviour of the bees.
4. Close observation of the hives condition has to be maintained. The colony should be at its peak strength at the time of nectar flow.
5. If more than one queen is formed in the colony it must be removed, otherwise it will lead to disintegration of the whole colony. Swarming must be prevented by removing excess drones and additional female queens. In cases the queen dies, immediately either a new queen is introduced or the whole member of the hive are transferred to another stock.

6. When there is dearth of food, artificial feeding may be resorted to. Artificial food is prepared by dissolving 100 gm of sugar in 150 ml of hot water, boiled and cooled. This food is kept near the hive in an open container. Sometimes during hot season, the food may get heated up. Water is required to keep the hive cool. Water availability near to the hive must be ensured.
7. For propagation, the number of colonies may be increased by dividing the existing stock.
8. Regular inspection of the hive for honey secretion and inspection of foundation frames to check the colony settlement may be made.
9. Predators and parasites found around the hive are eliminated.
10. The colony must be always maintained in a healthy state. If bees catch diseases they may be treated with available drugs.

APPLIANCES FOR MODERN METHOD

1. Typical movable hive.
2. Queen excluder.
3. Honey extractor.
4. Uncapping knife.
5. Other equipments.

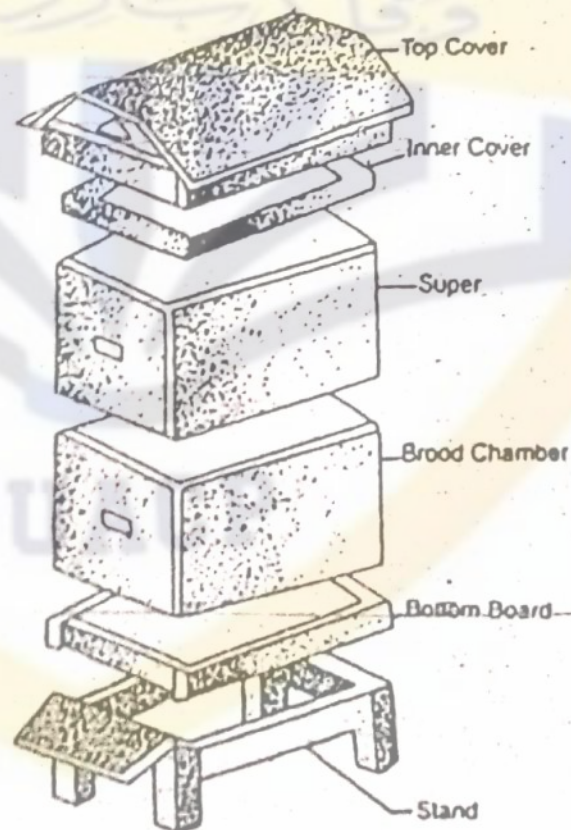


Fig. 3.7: Outer view of a two-storey langstroth-frame-hive.

1. Typical Movable Hive

An artificial movable hive is constructed by wooden box based on bee space theory. The size and number of frames are variable form hive to hive according to the need. A small space is enough to permit the entrance and exit of workers and drones but queen once placed in hive never comes outside the hive. The perforation size on zinc sheet is only of 0.375 cm but the thorax of the queen is 0.43 cm to 0.45 cm, so the queen can never pass through this pore. This typical hive consists of 6 parts as given below.

(a) Stand

It is the basal part of the hive on which the whole hive is constructed. The stands are adjusted to make slope for the hive. Due to this slope rainwater comes down quickly.

(b) Bottom board

It is situated above the stand and forms the proper base for the hive having two gates in the front position. One gate functions as an entrance while the other as exit.

(c) Brood chamber

The bottom board carries the brood chamber, which is the most important part of the bee hive. It is large in size provided with 5 to 10 frames. In each frame a wax sheet bearing hexagonal frames is held up by a couple of wires in a vertical position. Along with the margin of every hexagonal mark, the bees start making wall and ultimately the

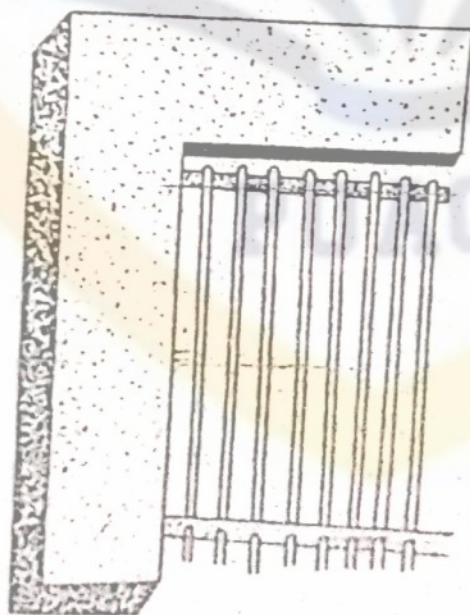


Fig. 3.8. Queen excluder (Waldron).

cells. Here every sheet of the wax is known as *comb foundation*, which attracts the bees and provides the base for the comb-preparation on both the sides. The frames are kept vertically in brood chamber, which is covered over by other frames having a wire meshing through which the workers can easily pass. The comb foundation helps in obtaining a regular strong worker brood cell comb, which can be used repeatedly. The Central Bee research station at Pune arranged the manufacture of a comb foundation mill, which manufactures, different cell sizes required in several regions of the country. The brood chamber is covered by another chamber known as super.

(d) Super

It is also without cover and the base. Super is provided with many frames containing comb foundation to provide additional space for expansion of the hive.

(e) Inner cover

It is a wooden piece used for the covering of the super. It has many holes for proper ventilation.

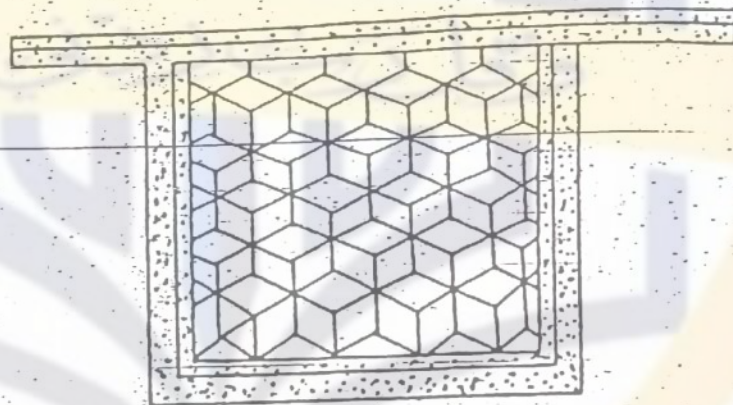


Fig. 3.9. Comb foundation fixed in a frame.

(f) Top cover

It is meant for protecting the colony from rains. It is fitted with zinc sheet, which is plain and sloping.

2. Queen Excluder

It consists for a wire gauze, extraneous guards and drone traps with individual wires placed 0.375 cm apart. It readily permits the workers to pass through it but keeps back the queen in the brood chamber.

3. Honey Extractor

It is used for the extraction of the honey from the comb and function on principle of centrifugal force. When combs are centrifuged



Fig. 3.10. Bee veil.

by his device the pure honey is thrown out without any damage to the comb.

4. Uncapping Knife

When all of the combs are filled with honey they are sealed by capping with the wax. So, before such capped combs are placed in the honey extractor, the wax sealing has to be removed with the help of an uncapping knife heated by steam before use.



Fig. 3.11. Uncapping knife.

5. Other Equipments

Most of the useful equipments for the successful management of the bee are locally manufactured which are very cheap. As they are made locally, they may not be exactly similar to those made at other place. Thus, Indian standard Institute has standardized some very common equipment. For the production of uniform and interchangeable articles. Some materials like protective garments, gum cages, gloves, net veil, bee net, brush etc. are required for easy and well planned handling of the bees.



Fig. 3.12. Bingham smoker.

ADVANCES OF MODERN METHOD

In the modern method of bee keeping there are several advantage which encourage the well planned bee keeping.

1. A proper watch on the activities of the bees can be had.
2. A strong colony can be developed by providing sugar, syrup, pollen substances to honey bees.
3. Swarming of bees is checked by modern hive.
4. The same hive is used again and again so the workers pay their attention more for the honey and not for the hive formation.
5. Under adverse climatic conditions the hive can be transferred from one place to the other for the protection of the bees.
6. Comb can be protected from the enemies.
7. Pure honey in large quantity can be obtained.

Precautions

For the proper management of bee keeping programme following precautions should be taken.

1. The hive should not be kept more than half a mile away from the place from where the bees have to collect the nectar and the pollen.
2. People must know about the bee keeper for proper contact.
3. The boxes must be kept under shade at cool places.



Fig. 3.13. Hive tool.

4. Industry should be near the road for proper transport facilities.
5. Fresh water reservoir should be near the hive.
6. Good flora for the collection of pollen and nectar should be there.

Economic Importance of Honey

Honey is used by human beings in different ways of which the most important is as food and medicine.

(a) Food value

It is estimated that 200 gm of honey provides as much nourishment as 11.5 litre of milk or 1.6 kg cream or 330 gm meat. 2.1 gm of honey provide as much as 67 K. cal of energy. Its sugars, minerals, vitamins and other vital elements are readily absorbed by the systems. Honey may be taken by healthy men as well as those who are ill. It can be taken at any time in any season and by persons of all ages even those just born. It is use in the preparation of candles, cakes and bread. In illness it is preferred over milk because more than half of the body energy is provided burning of dextrose.

(b) Medicinal value

Honey is mildly laxative, antiseptic and sedative, generally used in Ayurvedic and Unani systems of medicine. It is quite helpful in building up of the haemoglobin of the blood and also used as preventive against cough, cold and fever, as blood purifier and as a curative for ulcers on tongue and alimentary canal. Its regular use is recommended after severe cases of heart attack for malnutrition, indigestion and diabetes. It is also found that typhoid germs are killed by honey within 48 hours, those of branchio-pneumonia in 4 day and of dysentery in 50 hours.

(c) Other uses

Other than food and medicine, honey is used in numerous ways. It is used in the preparation of bread, cake and biscuits. It enhances their preserving quality. Much amount of honey goes in making alcoholic drinks. In poultry and fishing industries honey is widely used. In laboratory, honey is used to stimulate the growth of plants.

The bacterial culture, inoculation of seeds of cloves, in insect diet and in the preparation of poison baits for fruit flies.

Economic Importance of Beeswax

Beeswax is used in the manufacture of cosmetics, for Catholic churches, face cream, paints, ointments, insulators, plastic works, polishes, carbon paper and many other lubricant. It is also used in the laboratory for microtomy with the common wax for block preparation of tissues.

Bee Enemies

Enemies of the bees harm the colony in different ways so they have attracted considerable attention in the different regions of the country. The wax moths (*Galleria mellonella* and *Achroia grisella*), Wasp (*Vespa* spp. And *Palaris* sp.) black ants (*Componotus compressus*) and bee eaters (*Merops orientalis*) and Kingcrow (*Dicranus macrocerus*) are common enemies of the honey bee's comb and honey. Man is the last but worst enemy of honey bees.

Before 1958 bees were considered to be free from the diseases though suspected cases of *Nosema* from Punjab and Kashmir were known. But a parasitic mite *Acarapis woodi* Rennie caused *Acarine* disease in the adult honeybee in Kulu valley in Punjab in 1956. It was later reported from Himachal Pradesh, Uttar Pradesh and Jammu and Kashmir. This disease was controlled by the scheme in co-operation with the United States of America at the college of Agriculture Ludhiana, Punjab. Now-a-days Indian honey bees are continuously the spread of any such disease. A strict quarantine measure is being taken to check the spread of any disease from foreign countries. But in European countries bees are commonly attacked by microsporidian, which is injurious to bees.

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